

Multiple Source Seeking using Glowworm Swarm Optimization and Distributed Gradient Estimation

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Abstract—In this paper we propose a novel mission control strategy for a group of agents to locate unknown multiple extrema of a time-invariant scalar field. The goal is to provide a complete solution for realistic scenarios such as malfunction of agents or time-varying topology. Here we present MESA - Multiple Extrema Seeking Algorithm, where a group of autonomous agents are able, from any arbitrary initial location (in a bounded area), to locate multiple, unknown extrema. The proposed algorithm includes a combination of bio-inspired attraction with estimated gradient and formation control, where the task switching methodology is performed under the specification of Linear Temporal Logic formula. At first, agents find neighbors and construct groups for completing the search procedure. Once an extremum is "located", a virtual repulsive force is produced by the pioneer group which as a result cause others to search for other unexplored extrema. We propose a new measure-source density - evaluating the algorithm efficiency which is tested under different simulations for a unicycle agent's model.

I. INTRODUCTION

Performing a complex task such as exploring local extrema of a scalar field (e.g. concentration of oil spill or toxic cloud) using Multi-Agent System (MAS), requires intelligent algorithms which possess hybrid mission control characteristics. In this work we are following the growing line of bio-inspired approaches imitating natural swarm behavior. Such optimizations were proposed by many researches (see e.g. [1]), or for instance [2]–[4], where the Ant Colony Optimization (ACO) is introduced that uses shared information between agents regarding visited sites/nodes to find an optimal solution for the shortest route. Particle Swarm Optimization (PSO) is a population-based stochastic technique first introduced by [5] that can be used for various tasks such as source seeking and power systems [6]–[8]. Our proposed algorithm is based on the work conducted by Krishnanand and Ghose [9] regarding the Glowworm Swarm Optimization (GSO) technique. Similar to glowworms in nature, the agents possess a luminescence quantity named *luciferin* which defines a level of "attraction" to other agents. Solutions for different problems based on the GSO technique were proposed [10]–[12] in the past several years. For instance in [13], a sensor deployment scheme based on GSO is presented for better coverage to increase the effectiveness of a wireless sensor network. While the GSO involves restrictive assumptions such as initial area coverage, connectivity requirements (strongly depends on agents' "natural" sensing radius) and lack of robustness to time-varying field and agent dynamics, here

we use a combination of gradient climbing and formation control to propose an efficient approach. While a single-isolated maximum is being investigated thoroughly [14]–[16], the problem of finding multiple extrema has received little attention. Previous work [17] focused on providing a solution for exploring a time-varying field in a complex environment in terms of source location and level curve tracking. Here we wish to focus on the task of field exploration under realistic assumptions such as agent malfunction. This requires the algorithm to be robust to communication limits and varying connections between agents. For providing a formal specification for the hybrid control scheme, the Linear Temporal Logic (LTL) framework is used. LTL provides an intuitive formalization for expressing complex behavior such as task switching or temporal processes. There has been growing use of LTL over the past following years in diverse areas. Task planning under constraints [18], [19], motion planning [20], [21] and control synthesis [22] are only a few examples. For instance, in [23], a framework for generating hybrid controllers is proposed for performing high level tasks or behavior under satisfaction of different LTL formulas. The authors present solutions for different tasks (e.g., locating a fish, animal herding) with a single or a group of robots all specified and constrained by LTL formulas. Here, similar to [24], we formulate a control switching procedure in the form of LTL specifications to guaranty convergence and robustness.

The rest of the paper is organized as follows. In section II the problem statement is introduced. Section III describes the mission control algorithm (MESA) using LTL, GSO, formation control and gradient estimation. Section IV presents simulation results for a group of unicycle agents. Finally, in section V conclusions are drawn.

II. PROBLEM STATEMENT

Consider a multi-agent system of N mobile, autonomous agents and let $G = (V, E)$ be an undirected, connected graph where the communication topology is represented by the adjacency matrix

$$\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$$

$$a_{i,j} = \begin{cases} 1, & (i, j) \in E \\ 0, & \text{else} \end{cases} \quad (1)$$

or by the Laplacian matrix $L = \Delta - \mathcal{A}$, where $\Delta = \text{diag}(\mathcal{A} \cdot \mathbf{1})$ is the degree matrix of G with diagonal elements of $d_i = \sum_j a_{ij}$ and $\mathbf{1} = [1, \dots, 1]^T \in \mathbb{R}^N$. The proposed strategy is compatible with different agent models. In this paper we

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consider each agent as a non-holonomic unicycle with the kinematic model

$$\begin{aligned}\dot{x}_i(t) &= v_i \cos(\theta_i(t)) \\ \dot{y}_i(t) &= v_i \sin(\theta_i(t)) \\ \dot{\theta}_i(t) &= \omega_i(t),\end{aligned}\quad (2)$$

where, with respect to agent i , $r_i = [x_i, y_i] \in \mathbb{R}^2$ is the position in the plane, $\theta_i \in [0, 2\pi]$ is the orientation with respect to the x axis and $v_i, \omega_i \in \mathbb{R}$ are the control input for linear and angular velocities respectively. Let $\psi(t, r_i) : \mathbb{R}^+ \times \mathbb{R}^2 \rightarrow \mathbb{R}^+$ describe the time-varying level of a field at r_i .

Assumption 1: The initial search area is bounded.

The number of extrema (sources) in a field is N_ψ and $\hat{r} = [\hat{r}_1^T, \dots, \hat{r}_{N_\psi}^T]^T$ is their position vector. Field exploration requires agents to form groups where, a) δ is the maximum allowed number of agents in a single group; b) $\bar{r}_f$ is the offset between agents and the formation center of mass. In order to form such groups, a set

$$\begin{aligned}Q &= \{q_1, \dots, q_m\}, q_i \in \mathbb{R}^{2 \times m} \\ \mathcal{Q} &= \{1, 2, \dots, m\} \\ \min \|q_i - q_j\| &\geq 2r_s, \forall i, j \in \mathcal{Q}\end{aligned}\quad (3)$$

of m predefined locations, acting as "match-maker", is generated,

$$m = \left\lceil \frac{h_f w_f}{[2r_s]^2} \right\rceil, \quad (4)$$

where h_f and w_f is the field maximum height and width respectively, and r_s is an agent's maximum sensing radius. The number of extrema expected to be detected is decided by $\tilde{N}_\psi$, which determines the number of agents

$$N = \delta \cdot \tilde{N}_\psi. \quad (5)$$

The task splits to two parts: locating a single source and swarm field exploration. For the first part, consider $\psi_\alpha = \psi(\hat{r}_\alpha)$, $\alpha \in [1, N_\psi]$ as the concentration level of a source located in $\hat{r}_\alpha$.

Definition 1: The set of all agents which are locating an extremum at $\hat{r}_\alpha$ is defined as

$$\xi_\alpha = \{i \in N \mid \|\hat{r}_\alpha - r_i\| \leq \bar{r}_f\}. \quad (6)$$

Now, for the second part, for evaluating the field exploration performance of a swarm, we define the **source density**.

Definition 2: A source density is defined as

$$p_{\psi_\alpha} = \frac{1}{N} \sum_{i \in \xi_\alpha} e^{-\|\hat{r}_\alpha - r_i\|}, \quad (7)$$

where $p_{\psi_\alpha} \in [0, 1]$, $\alpha \in [1, N_\psi]$.

From (7), if $p_{\psi_\alpha} = 1$ then all agents are at the extremum location $\hat{r}_\alpha$. For the case of multiple extrema it is required that all densities shall be leveled (i.e., $p_{\psi_1} = \dots = p_{\psi_{N_\psi}}$).

Given a field, and under the assumption that the search area boundaries are known, the problem is to design a mission control strategy where; a) the agents are able to locate and track multiple extrema. b) the agents are able to

locate all extrema with equal density. The agents, based only on local information, will have to establish communication with near neighbors and reach consensus regarding formation and source location. For a more realistic approach we allow agents to malfunction, where the swarm has to react accordingly to still achieve the task.

For such a mission control design, we use the specification of an LTL formula φ to control the switching between tasks. Here we use a fragment of LTL, where the formula is constructed over a set of atomic propositions ($\{a\} \in AP$) according to the grammar

$$\varphi := \text{true} \mid a \mid \varphi_1 \wedge \varphi_2 \mid \neg \varphi \mid \bigcirc \varphi \mid \varphi_1 \mathcal{U} \varphi_2, \quad (8)$$

equipped with traditional ($\wedge, \vee, \neg$) and temporal ($\bigcirc, \Diamond, \bigcirc, \mathcal{U}, \rightarrow$) operators. We use the standard definitions and semantics of LTL as in [24], [25].

III. MULTIPLE EXTREMA SEEKING ALGORITHM

In this section we present MESA - a mission control strategy for the described problem. The idea is that after random deployment of the agents in a bounded area covered by equally distributed predefined match-making points, groups are formed while approaching a nearest match-making location, and search the plane for extrema using gradient climbing and attraction forces. Once an extremum is located, the respective group generates a virtual repulsive force for marking the location to encourage other agents to search for other extremum.

A. Source Seeking

First, we describe the control design for locating an extremum. Our approach is based on a combination of different elements such as distributed gradient and luciferin level while keeping formation. A key element in the search procedure is the sensing radius r_s . This way connections are made and information is shared. The GSO method [9] is based on adjusting agent attractiveness. Let l_i be a property which defines the attraction level of agent i , referred to as *luciferin* and given by

$$l_i(r_i, t) = (1 - \rho)l_i(t - \Delta t) + \zeta|\psi(r_i(t))|, \quad (9)$$

where ρ, ζ are pre-defined tuning parameters representing the luciferin decay ($0 < \rho < 1$) and enhancement constants and Δt is the time step interval. Each agent has a dynamic attraction radius r_d^i that limits the attraction area and is determined as

$$r_d^i(t + \Delta t) = \min\{r_s, \max\{0, r_d^i(t) + \kappa(\delta - |\mathcal{N}_i(t)|)\}\}, \quad (10)$$

where $\kappa > 0$ is a constant parameter. Then, the attraction force of agent j exerted on i is

$$p_{ij} = \frac{l_j - l_i}{\sum_{k \in \tilde{\mathcal{N}}_i} l_k - l_i}, \quad (11)$$

where $j \in \tilde{\mathcal{N}}_i$, $\tilde{\mathcal{N}}_i = \{j : d_{ij} < r_d^i; l_i < l_j\}$ is the set of neighbors of agent i and d_{ij} is the Euclidean distance.

Eventually, agent i is attracted to its most attractive neighbor $\{\tilde{j} = j \mid p_{ij} = \max_j(p_{ij})\}$, then

$$e_i^l(t) = s \frac{r_{\tilde{j}}(t) - r_i(t)}{\|r_{\tilde{j}}(t) - r_i(t)\|}, \quad (12)$$

where $e_i^l \in \mathbb{R}^2$ and $s > 0$ is a tuning parameter.

For moving toward possible extrema or reacting to variations of the field, we implement a distributed gradient estimation [26]. Here, the agent's neighbors are defined according to the sensing radius r_s and under the limitation of at most δ agents in a group. By computing the slope between agent i and $j \in \mathcal{N}_i$ we obtain

$$\underbrace{\begin{bmatrix} \psi(r_{i_1}) - \psi(r_i) \\ \vdots \\ \psi(r_{i_{|\mathcal{N}_i|}}) - \psi(r_i) \end{bmatrix}}_{b_i} = \underbrace{\begin{bmatrix} (r_{i_1} - r_i)^T \\ \vdots \\ (r_{i_{|\mathcal{N}_i|}} - r_i)^T \end{bmatrix}}_{R_i} \nabla \psi(\bar{r}_i), \quad (13)$$

where $\bar{r}_i(t) = [r_{i_1}(t), \dots, r_{i_{|\mathcal{N}_i|}}(t)]^T$, $b_i \in \mathbb{R}^{|\mathcal{N}_i|}$, $R_i \in \mathbb{R}^{|\mathcal{N}_i| \times 2}$. Using least-squares we obtain the distributed estimated gradient

$$\hat{g}_i(t) = (R_i^T R_i)^{-1} R_i^T b_i. \quad (14)$$

The inverse of $(R_i^T R_i)^{-1}$ exists iff R_i is full column rank, which requires $2 \leq |\mathcal{N}_i|$ or $\delta \geq 3$. Then,

$$e_i^g(t) = K_g \cdot |\hat{g}_i(t)| \cdot \text{sign}(e_i^l(t)), \quad (15)$$

where $e_i^g, K_g \in \mathbb{R}^2$ and K_g is a tuning parameter.

For estimation of the gradient by taking measurement in different locations, each group arranges itself in a specified formation

$$e^f(t) = K_f L_p (r^f - r(t)), \quad (16)$$

where $K_f \in \mathbb{R}^2$ is a tuning parameter, r^f is the relative formation vector and $L_p = L \otimes I_p \in \mathbb{R}^{pN \times pN}$ (here $p = 2$).

B. Field Exploration

Now we describe the second part concerning the field exploration performance of a swarm. Each agent carries a predefined set of reference locations according to (3),(4). This set's contribution is twofold; neighbors "match-maker" and extremum density. While an agent is not a part of a group it keeps searching for neighbors by traveling between the reference locations according to

$$\begin{aligned} q_p^i(t) &= \arg \min_{p \in \mathcal{J}_Q} \|q_p - r_i(t)\| \\ e_i^q(t) &= K_q \frac{q_p^i(t) - r_i(t)}{\|q_p^i(t) - r_i(t)\|} \end{aligned} \quad (17)$$

where $K_q > 0$ is a tuning parameter. If an agent reaches q_p without finding a neighbor, then this location is excluded and the search continues to the next nearest reference location $q_b \in Q$, $b \neq p \in \mathcal{J}_Q$. The same is applicable for locating all possible extrema. Once a group locates an extremum ψ_α , if an other agent i approaches, and

$$\|r_j - r_i\| \leq r_s, j \in \xi_\alpha, i \notin \mathcal{N}_j \quad (18)$$

then a virtual repulsive force is generated by compelling the approaching agents to search for the nearest reference location according to (17) and the searching procedure is repeated.

The composition of all with respect to agent i ,

$$e_i(t) = \rho_i^q e_i^q(t) + \rho_i^l e_i^l(t) + \rho_i^g e_i^g(t) + \rho_i^f e_i^f(t), \quad (19)$$

where $e_i \in \mathbb{R}^2$, $e = [e_1, \dots, e_N]^T \in \mathbb{R}^{2N}$ and

$$\begin{aligned} \rho_i^q &= \begin{cases} 0, |\mathcal{N}_i| = \delta \wedge \|\hat{g}\| > 0 \\ 1, \text{ else} \end{cases}; \rho_i^l = \begin{cases} 0, (18) \text{ holds} \\ 1, \text{ else} \end{cases} \\ \rho_i^g &= \begin{cases} 0, |\mathcal{N}_i| < 2, (18) \text{ holds} \\ 1, \text{ else} \end{cases}; \rho_i^f = \begin{cases} 0, |\mathcal{N}_i| < 1 \\ 1, \text{ else} \end{cases}, \end{aligned} \quad (20)$$

where $\delta = \delta - 1$.

The local control law which stabilizes and navigates the non-holonomic unicycle is similar to [27]

$$\begin{aligned} u_i(t) &= [v_i(t), \omega_i(t)]^T \\ v_i(t) &= k_v D_i(t) \cos(e_{\theta_i}(t)) \\ \omega_i(t) &= k_\omega e_{\theta_i}(t) + \dot{\theta}_{d_i}(t), \end{aligned} \quad (21)$$

where $e_i(t) = [e_{x_i}, e_{y_i}, e_{\theta_i}]^T$, $D_i = \sqrt{e_{x_i}^2 + e_{y_i}^2}$, $e_{\theta_i} = \theta_{d_i} - \theta_i$, $\theta_{d_i} = \tan^{-1}(\frac{e_{y_i}}{e_{x_i}})$ and $k_\omega, k_v > 0$ are tuning parameters.

Assumption 2: It is assumed that $e_{\theta_i} \neq k\frac{\pi}{2}$, $k \in \{-1, -3, 1, 3\}$ for all agents; In other words $\cos(e_{\theta_i}) \neq 0, \forall i \in N$. When $e_{\theta_i} = k\frac{\pi}{2}$, the orientation error should be perturbed to $e_{\theta_i} = e_{\theta_i} + \varepsilon_\theta$

The complete control scheme is presented in Figure 1.

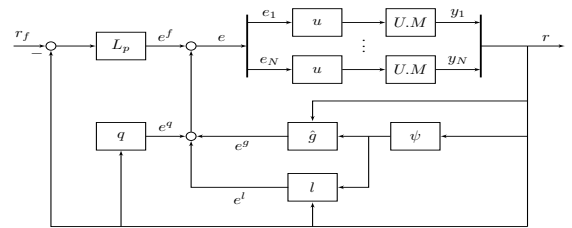


Fig. 1: Proposed distributed control scheme

C. LTL Specification

Here we formalize MESA using a linear temporal logic specification formula. We use (20) to define our atomic proposition set $\{a_1, a_2, a_3, a_4\} \in AP$

$$\begin{aligned} a_1 &: |\mathcal{N}_i| < \delta \\ a_2 &: |\mathcal{N}_i| = \delta \\ a_3 &: \|e_i\| \leq \|K_g |\hat{g}_i(\bar{r}^f)| + s\| \\ a_4 &: \|r_j - r_i\| \leq r_s, j \in \xi_\alpha, i \notin \mathcal{N}_j \\ a_5 &: \|q_b - r_i\| > 0. \end{aligned} \quad (22)$$

When a_1 is true then the searching neighbors and forming groups procedure is active which is completed once a_2

is *true*. a_3 is *true* when an agent is near an un-located extremum and a_4 is *true* when agent i approaches occupied extremum which leads to a_5 and a repulsive force. This can be stated as an LTL specification formula

$$\varphi = \square[(a_1 \rightarrow \Diamond a_2) \mathcal{U} a_3 \wedge (a_4 \bigcirc a_5 \rightarrow \Diamond a_1)]. \quad (23)$$

Next we build a relevant Buchi Automaton [28], [29], $\mathcal{A}_\varphi$, which accepts this specification and consists of four states (Figure 2):

- b_0 : $\rho_i^q, \rho_i^l, \rho_i^f, \rho_{i,|\mathcal{N}_i| \geq 1}^g, \rho_{i,|\mathcal{N}_i| \geq 2}^g = 1$
- b_1 : $\rho_i^l, \rho_i^f, \rho_i^g = 1, \rho_i^q = 0$
- b_2 : $i \in \xi_\alpha$
- b_3 : $\rho_i^l, \rho_i^f, \rho_i^g = 0, \rho_i^q = 1$

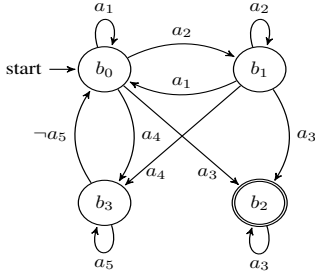


Fig. 2: Automaton $\mathcal{A}_\varphi$

From $\mathcal{A}_\varphi$ one can see that the system starts in b_0 where the relevant ρ_i is active for connecting with neighbors; once a full group is completed the system switches to b_1 where ρ_i^q is canceled. Approaching a source which is already "occupied" by other groups causes transition to state b_3 where only ρ_i^q is active and the agent is rejected from this extremum. Once reaching the reference location, it returns to b_0 and the procedure is repeated searching for a different extremum. b_2 is defined as the accepting state where an "unoccupied" extremum is located; it can be reached from both b_0 and b_1 . If $\tilde{N}_\psi = N_\psi$ then all extrema are occupied, each with δ agents, all ending in state b_2 . Otherwise, if $\tilde{N}_\psi < N_\psi$ then only $\tilde{N}_\psi$ extrema are located. If $\tilde{N}_\psi > N_\psi$ then all N_ψ extrema are located while $\delta(\frac{N}{\delta} - N_\psi)$ agents continue to move around the bounded search area.

D. Convergence Analysis

Next we analysis convergence under the LTL specification.

Proposition 1: Consider the multi-agent system (2) with the control law (21) and suppose that Assumptions 1,2 are fulfilled. Then, under the LTL formula specification (23), for any $r(0) \in \mathbb{R}^{2N}$ and $t \geq 0$, the agents locate $\min(N_\psi, \lfloor \frac{N}{\delta} \rfloor)$ unknown multiple extrema with a density error of $e_{\psi_\alpha} = \frac{1}{N_\psi} \sum_{\beta=1}^{N_\psi} (p_{\psi_\alpha} - p_{\psi_\beta})^2 \leq \varepsilon_\kappa$, $\alpha \in [1, N_\psi]$, and a position error (19) of $e_i \leq \|K_g \hat{g}_i(\bar{r}^f)\| + s$, $\forall i \in N$.

Proof: First we prove a group converges to a single extremum Similar to [27], consider the Lyapunov candidate function

$$V(x, y) = \frac{1}{2}(e_{x_i}^2 + e_{y_i}^2 + e_{\theta_i}^2) \quad (24)$$

Then,

$$\dot{V} = -\dot{x}_i e_{x_i} - \dot{y}_i e_{y_i} - k_\omega e_{\theta_i}^2. \quad (25)$$

Now, by substituting $\theta_i = e_{\theta_i} + \theta_{d_i}$ and from (21) using $v_i = k_v D_i \cos(e_{\theta_i})$ and $\cos(\theta_{d_i}) = \frac{e_{x_i}}{D_i}$, $\sin(\theta_{d_i}) = \frac{e_{y_i}}{D_i}$ in (2) we obtain

$$\dot{V} = -k_v(e_{x_i}^2 + e_{y_i}^2)\cos^2(e_{\theta_i}) - k_\omega e_{\theta_i}^2 \quad (26)$$

hence, $\dot{V} \leq 0$ and $\dot{V} = 0$ if $e_{x_i} = e_{y_i} = e_{\theta_i} = 0$; thus the desired formation is achieved.

According to [9], when formation is not applied, the luciferin level asymptotically converges to $l_i^* = (\frac{\xi}{\rho})\psi(r_i)$. By considering formation, agents cannot moving independently but sharing attractive forces which eventually leads to

$$\lim_{t \rightarrow \infty} e_i^l(t) = \lim_{t \rightarrow \infty} s \frac{r_j^*(t) - r_i(t)}{\|r_j^*(t) - r_i(t)\|} \leq s \cdot \frac{\bar{r}^f}{\bar{r}^f} = s, \quad (27)$$

where $\bar{r}^f$ is the offset between agents and the formation center of mass. As a result it is also received that

$$\lim_{t \rightarrow \infty} e_i^g(t) = \lim_{t \rightarrow \infty} K_g \cdot |\hat{g}_i(t)| \cdot \text{sign}(e_i^l(t)) \leq K_g |\hat{g}_i(\bar{r}^f)|. \quad (28)$$

Observing the LTL formula one can see that once an extremum is located then the accepting state b_2 is reached and the agent remains there. In other words, MESA terms and conditions are guided by the LTL specification.

Now, we can prove the density result. According to Definition 1, a group is defined as belonging to an extremum ψ_α as $\xi_\alpha = \{i \in N \mid \|\hat{r}_\alpha - r_i\| \leq \bar{r}^f\}$. Then, the source density is (according to Definition 2)

$$p_{\psi_\alpha} = \frac{1}{N} \sum_{i \in \xi_\alpha} e^{-\|\hat{r}_\alpha - r_i\|} \leq \frac{1}{N} \sum_{i \in \xi_\alpha} e^{-\bar{r}^f} = \frac{\delta}{N} e^{-\bar{r}^f}. \quad (29)$$

This is true $\forall \alpha \in [1, \tilde{N}_\psi]$, i.e., $p_{\psi_\alpha} = p_{\psi_\beta}$, $\forall \alpha, \beta \in [1, \tilde{N}_\psi]$. So the density error between the different extrema for the case of $\tilde{N}_\psi \geq N_\psi$ is

$$e_{\psi_\alpha} = \frac{1}{N_\psi} \sum_{\beta=1}^{N_\psi} (p_{\psi_\alpha} - p_{\psi_\beta})^2 = 0. \quad (30)$$

We need also look at the case when $\tilde{N}_\psi < N_\psi$ and show that the error is bounded,

$$\begin{aligned} e_{\psi_\alpha} &= \frac{1}{N_\psi} \sum_{\beta=1}^{N_\psi} (p_{\psi_\alpha} - p_{\psi_\beta})^2 \\ &= \frac{1}{N_\psi} \sum_{\beta=1}^{\tilde{N}_\psi} (p_{\psi_\alpha} - p_{\psi_\beta})^2 + \frac{1}{N_\psi} \sum_{\beta=\tilde{N}_\psi+1}^{N_\psi} (p_{\psi_\alpha} - p_{\psi_\beta})^2. \end{aligned} \quad (31)$$

We distinguish two cases. When $\psi_\alpha \in [1, \tilde{N}_\psi]$ then as before, $p_{\psi_\alpha} = p_{\psi_\beta} = \frac{\delta}{N} e^{-\bar{r}^f}$, $\forall \alpha, \beta \in [1, \tilde{N}_\psi]$ and $\forall \beta \in [\tilde{N}_\psi + 1, N_\psi]$, $p_{\psi_\beta} = 0$ so

$$\begin{aligned} e_{\psi_\alpha \in [1, \tilde{N}_\psi]} &= \frac{1}{N_\psi} \sum_{\beta=\tilde{N}_\psi+1}^{N_\psi} \left(\frac{\delta}{N} e^{-\bar{r}^f} \right)^2 \\ &= \underbrace{\frac{N_\psi - \tilde{N}_\psi}{N_\psi} \left(\frac{\delta}{N} e^{-\bar{r}^f} \right)^2}_{\kappa_1} \leq \varepsilon_\kappa \kappa_1, \end{aligned} \quad (32)$$

where $\varepsilon_\kappa > 1$. When $\alpha \notin [1, \tilde{N}_\psi]$ then $p_{\psi_\alpha} = 0$ and

$$\begin{aligned} e_{\psi_\alpha \in [\tilde{N}_\psi+1, N_\psi]} &= \frac{1}{N_\psi} \sum_{\beta=1}^{\tilde{N}_\psi} \left(-\frac{\delta}{N} e^{-\bar{r}_f}\right)^2 \\ &= \underbrace{\frac{\tilde{N}_\psi}{N_\psi} \left(\frac{\delta}{N} e^{-\bar{r}_f}\right)^2}_{\kappa_2} \leq \varepsilon_\kappa \kappa_2. \end{aligned} \quad (33)$$

IV. SIMULATION RESULTS

In this section we present simulation results for exploring a scalar field with 5 extrema with different number of unicycle agents constructing groups of $\delta = 4$. The time-invariant scalar field used here is the *Peaks* function

$$\begin{aligned} \psi(r) &= \eta_1(1-x)^2 e^{-(x^2+(y+1)^2)} \\ &- \eta_2\left(\frac{x}{5} - x^3 - y^5\right) e^{-(x^2+y^2)} - \eta_3 e^{-((x+1)^2+y^2)}, \end{aligned} \quad (34)$$

where $\eta = [3, 10, 1/3]$. The field is constructed of one global maxima in $\psi_1 = (-0.01, 1.6)$ and two local ones in $\psi_2 = (-0.46, -0.63)$, $\psi_3 = (1.28, 0)$ in addition to one global minima in $\psi_4 = (0.24, -1.65)$ and a local one in $\psi_5 = (-1.35, 0.2)$. First, we study the case where $\tilde{N}_\psi = 5$ and with groups of $\delta = 4$, which according to (5), the number of agents for this task is taken to be $N = 20$. Figure 3 shows simulation results in the $x-y$ plane. At the beginning, the agents (gray rectangle/black top circle) are distributed arbitrary in the bounded search area (b). Then, by moving toward the reference locations (red-cross marks) the agents locate neighbors (state b_0) and cluster into groups while detecting attractive agents and gradient changes. Then (a_2 is true change to state b_1), the agents climb or descent toward an extremum until each group is settled and the mission is completed (c), i.e., a_3 is true, change to the acceptance state b_2 . If a single agent or a group approaches an occupied extremum, then they repulse and navigate to the nearest reference location (i.e., state b_3) until reaching it and repeat the search process (state b_0). The agents' trajectories during the whole process in time are shown in (a).

Table I shows a comparison of the sources densities according to (7) between MESA and distributed gradient climbing (similar to [26] for the same set of initial conditions). One can see that the gradient method, which is

		ψ_1	ψ_2	ψ_3	ψ_4	ψ_5
MESA	p_{ψ_α}	0.1665	0.1654	0.1656	0.1666	0.1656
	eq.(29)	0.1637	0.1637	0.1637	0.1637	0.1637
	e_{ψ_α}			$< 1e^{-6}$		
Gradient	p_{ψ_α}	0.2459	0.3268	0.1655	0	0
	e_{ψ_α}	0.0268	0.0492	0.0175	0.0389	0.0389
	Avg. p_{ψ}					
MESA		0.1659	0.0006	0.0000	0.1666	
Gradient		0.1476	0.1463	0.0343	0.3268	

TABLE I: Performances comparison between different methods, $\tilde{N}_\psi = N_\psi = 5$, $N = 20$, $\delta = 4$

satisfactory for locating a single isolated extremum, fails in

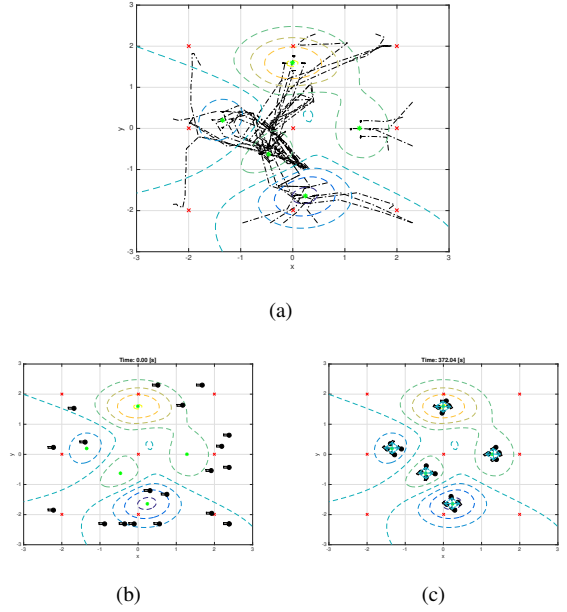


Fig. 3: The swarm exploration behavior in a $x-y$ plot: (a) agents' trajectories from initial locations to the extrema (b) initial arbitrary distribution of the agents (c) final location. The agents possess each extremum in a desired diamond formation. See animated example: https://www.tuhh.de/t3resources/ics/user_upload/MESA.mp4

multiple seeking mission, while ψ_2 has the highest density, ψ_4, ψ_5 were not discovered. A different approach such as only GSO will not guaranty improvement. MESA shows equal source density levels and a low standard deviation. Moreover, the measured density is similar to the calculated one according to (29) and the error is almost zero.

Tables II, III present the results of two additional simulations one for $\tilde{N} = 4$, $N = 16$ and the other for $\tilde{N} = 6$, $N = 24$. For the case when $\tilde{N}_\psi < N_\psi$, from Table II one can

	ψ_1	ψ_2	ψ_3	ψ_4	ψ_5
p_{ψ_α}	0.2095	0.2074	0	0.2084	0.2067
e_{ψ_α}	0.0088	0.0086	0.0346	0.0087	0.0085
eq.(32)	0.0092	0.0092	-	0.0092	0.0092
eq.(33)	-	-	0.0368	-	-

TABLE II: For $\tilde{N}_\psi < N_\psi$, $N = 16$, and $\varepsilon_\kappa = 1.1$

	ψ_1	ψ_2	ψ_3	ψ_4	ψ_5
p_{ψ_α}	0.1388	0.1384	0.1380	0.1377	0.1377
eq.(29)	0.1365	0.1365	0.1365	0.1365	0.1365
e_{ψ_α}			$< 1e^{-6}$		

TABLE III: For $\tilde{N}_\psi > N_\psi$, $N = 24$

notice that ψ_3 was not detected due to the low number of agents. The other extrema were located with equal density. The measured value e_{ψ_α} is almost equal to the calculated one (32),(33) according to whether an extremum is "located" or not. For the case where $\tilde{N}_\psi > N_\psi$, Table III results are similar to $\tilde{N}_\psi = N_\psi$ with lower source density (due to the larger number of agents in use). While the first 20 agents locate and remain in all extrema locations, the redundant 4

agents keeps moving in the region, searching for potential field changes.

V. CONCLUSIONS

In this paper, MESA - a novel mission control strategy for exploring a scalar field is proposed. A design using a combination of Glowworm Swarm Optimization (GSO), distributed gradient and formation control is presented providing a solution for locating different extrema. The approach involves several states and conditions which are defined and controlled under the specification of a Linear Temporal Logic (LTL) formula. It has been shown that the proposed algorithm, under the assumption of a bounded search area and estimated number of extrema, guarantee exploration of multiple extrema with equal value of source density. An important feature of this approach is its robustness to realistic changes and disturbances such as agent malfunction. The algorithm performance is tested under different simulation scenarios and compared with the results of distributed gradient climbing. Subject of current research is the application of this approach for the exploration of time-varying fields under a realistic environments (oil spill) using Autonomous Underwater Vehicle.

REFERENCES

- [1] P. Leitão, J. Barbosa, and D. Trentesaux, "Bio-inspired multi-agent systems for reconfigurable manufacturing systems," *Engineering Applications of Artificial Intelligence*, vol. 25, no. 5, pp. 934 – 944, 2012. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0952197611001886>
- [2] M. Dorigo and G. D. Caro, "Ant colony optimization: a new meta-heuristic," in *Proceedings of the 1999 Congress on Evolutionary Computation-CEC99 (Cat. No. 99TH8406)*, vol. 2, 1999, p. 1477 Vol. 2.
- [3] M. Dorigo, M. Birattari, and T. Stutzle, "Ant colony optimization," *IEEE Computational Intelligence Magazine*, vol. 1, no. 4, pp. 28–39, Nov 2006.
- [4] W. Xiang and H. Lee, "Ant colony intelligence in multi-agent dynamic manufacturing scheduling," *Engineering Applications of Artificial Intelligence*, vol. 21, no. 1, pp. 73 – 85, 2008. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0952197607000437>
- [5] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Neural Networks, 1995. Proceedings., IEEE International Conference on*, vol. 4, Nov 1995, pp. 1942–1948 vol.4.
- [6] B. Zhao, C. X. Guo, and Y. J. Cao, "A multiagent-based particle swarm optimization approach for optimal reactive power dispatch," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 1070–1078, May 2005.
- [7] Y. del Valle, G. K. Venayagamoorthy, S. Mohagheghi, J. C. Hernandez, and R. G. Harley, "Particle swarm optimization: Basic concepts, variants and applications in power systems," *IEEE Transactions on Evolutionary Computation*, vol. 12, no. 2, pp. 171–195, April 2008.
- [8] R. Zou, V. Kalivarapu, E. Winer, J. Oliver, and S. Bhattacharya, "Particle swarm optimization-based source seeking," *IEEE Transactions on Automation Science and Engineering*, vol. 12, no. 3, pp. 865–875, July 2015.
- [9] K. N. Krishnanand and D. Ghose, "Glowworm swarm optimization for simultaneous capture of multiple local optima of multimodal functions," *Swarm Intelligence*, vol. 3, no. 2, pp. 87–124, 2009. [Online]. Available: <http://dx.doi.org/10.1007/s11721-008-0021-5>
- [10] B. Wu, C. Qian, W. Ni, and S. Fan, "The improvement of glowworm swarm optimization for continuous optimization problems," *Expert Systems with Applications*, vol. 39, no. 7, pp. 6335 – 6342, 2012. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0957417411016885>
- [11] Q. Gong, Y. Zhou, and Q. Luo, "Hybrid artificial glowworm swarm optimization algorithm for solving multi-dimensional knapsack problem," *Procedia Engineering*, vol. 15, pp. 2880 – 2884, 2011, {CEIS} 2011. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S1877705811020431>
- [12] L. He and S. Huang, "Improved glowworm swarm optimization algorithm for multilevel color image thresholding problem," *Mathematical Problems in Engineering*, vol. 2016, 2016.
- [13] W.-H. Liao, Y. Kao, and Y.-S. Li, "A sensor deployment approach using glowworm swarm optimization algorithm in wireless sensor networks," *Expert Systems with Applications*, vol. 38, no. 10, pp. 12 180 – 12 188, 2011. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0957417411004611>
- [14] C. Paliotta, D. J. Belleter, and K. Y. Pettersen, "Adaptive source seeking with leader-follower formation control," *IFAC-PapersOnLine*, vol. 48, no. 16, pp. 285–290, 2015.
- [15] E. Rosero and H. Werner, "Cooperative source seeking via gradient estimation and formation control (part 2)," in *UKACC International Conference on Control*, 2014.
- [16] J. Cochran, A. Siranosian, N. Ghods, and M. Krstic, "3-d source seeking for underactuated vehicles without position measurement," *IEEE Transactions on Robotics*, vol. 25, no. 1, pp. 117–129, Feb 2009.
- [17] A. Turgeman and H. Werner, "Mission control - combined solutions for source seeking and level curve tracking in a time-varying field," in *2017 American Control Conference (ACC)*, May 2017, pp. 4268–4273.
- [18] M. Kloetzer, X. C. Ding, and C. Belta, "Multi-robot deployment from LTL specifications with reduced communication," *CoRR*, vol. abs/1108.3240, 2011. [Online]. Available: <http://arxiv.org/abs/1108.3240>
- [19] Y. Kantaros and M. M. Zavlanos, "A distributed ltl-based approach for intermittent communication in mobile robot networks," in *2016 American Control Conference (ACC)*, July 2016, pp. 5557–5562.
- [20] Z. Zhang and R. V. Cowlagi, "Motion-planning with global temporal logic specifications for multiple nonholonomic robotic vehicles," in *2016 American Control Conference (ACC)*, July 2016, pp. 7098–7103.
- [21] M. Guo, K. H. Johansson, and D. V. Dimarogonas, "Revising motion planning under linear temporal logic specifications in partially known workspaces," in *Robotics and Automation (ICRA), 2013 IEEE International Conference on*. IEEE, 2013, pp. 5025–5032.
- [22] I. Papusha, J. Fu, U. Topcu, and R. M. Murray, "Automata theory meets approximate dynamic programming: Optimal control with temporal logic constraints," in *2016 IEEE 55th Conference on Decision and Control (CDC)*, Dec 2016, pp. 434–440.
- [23] G. E. Fainekos, H. Kress-Gazit, and G. J. Pappas, "Temporal logic motion planning for mobile robots," in *Proceedings of the 2005 IEEE International Conference on Robotics and Automation*, April 2005, pp. 2020–2025.
- [24] S. G. Loizou and K. J. Kyriakopoulos, "Automatic synthesis of multi-agent motion tasks based on ltl specifications," in *2004 43rd IEEE Conference on Decision and Control (CDC) (IEEE Cat. No.04CH37601)*, vol. 1, Dec 2004, pp. 153–158 Vol.1.
- [25] H. Kress-Gazit, G. E. Fainekos, and G. J. Pappas, "Temporal-logic-based reactive mission and motion planning," *IEEE Transactions on Robotics*, vol. 25, no. 6, pp. 1370–1381, Dec 2009.
- [26] E. Rosero and H. Werner, "Cooperative source seeking via gradient estimation and formation control (part 1)," in *UKACC International Conference on Control*, 2014.
- [27] S. Ahmadi Barogh, E. Rosero, and H. Werner, "Formation control of non-holonomic agents with collision avoidance," in *American Control Conference*, 2015.
- [28] P. Gastin and D. Oddoux, *Fast LTL to Büchi Automata Translation*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2001, pp. 53–65. [Online]. Available: http://dx.doi.org/10.1007/3-540-44585-4_6
- [29] P. Wolper, "Constructing automata from temporal logic formulas: A tutorial," in *Lectures on Formal Methods and Performance Analysis*, E. Brinksma, H. Hermanns, and J.-P. Katoen, Eds. New York, NY, USA: Springer-Verlag New York, Inc., 2002, pp. 261–277. [Online]. Available: <http://dl.acm.org/citation.cfm?id=567305.567312>